

Jayonto Dutta Plabon

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🌐 [jayonto-dutta-plabon](#) 🔗 [Plabon-Dutta](#)

Research interests: Large Language Models for Low-Resource Languages, Federated Learning, Explainable AI, and Privacy-Preserving Machine Learning.

🎓 Education

2024–Present Expected: Jul’26	M.Sc. in Computer Science and Engineering (Running) University of Asia Pacific, Bangladesh <i>Thesis: BanglaLex: A Fine-Tuned Large Language Model for Bangladeshi Legal Advisory with Retrieval-Augmented Generation and Hallucination Grounding (Expected Completion: June 2026) – using a domain-specific RAG pipeline and a hallucination grounding mechanism to suppress factually unreliable outputs</i>	CGPA: 4.00*/4.00
2020–2024	B.Sc. in Computer Science and Engineering University of Asia Pacific, Bangladesh <i>Thesis: Explainable federated learning for privacy-preserving bangla sign language detection — conducted a systematic evaluation of five ML models (CNN, MobileNet, ResNet, VGG19, Inception) across three federated aggregation algorithms (FedAvg, FedProx, FedOpt) for privacy-preserving Bangla sign language detection eliminating reliance on centralized biometric data.</i>	CGPA: 3.87/4.00

📁 Employment

Jul 25–Now	Lecturer Dept. of Computer Science and Engineering, University of Asia Pacific Teaching undergraduate courses including Computer Fundamentals (50+ students) and Competitive Programming (60+ students); mentoring 8+ capstone projects on federated learning and NLP applications.	Dhaka Bangladesh
Jul 24–Jan 25	Teaching Assistant Dept. of Computer Science and Engineering, University of Asia Pacific Served as TA for the OOP and DSA Lab; conducted weekly lab sessions for 60+ students, graded assignments, and provided one-on-one guidance during office hours.	Dhaka Bangladesh
Jan 24–May 24	Research Assistant Advance Machine Intelligence Research Lab Designed and implemented federated model training pipelines for emoji prediction tasks; contributed to experimental methodology and result analysis on published study (J2).	Dhaka Bangladesh

📄 Publication

Journal Articles

- 2024 J1 **Explainable federated learning for privacy-preserving bangla sign language detection**
BS Diba, JD Plabon, MDM Rahman, D Mistry, AK Saha, MF Mridha
Engineering Applications of Artificial Intelligence (Q1, Impact Factor: 8.0) – 20 citations
DOI: [10.1016/j.engappai.2024.108657](https://doi.org/10.1016/j.engappai.2024.108657) ; Implementation: 🔄
- J2 **Federated Learning-Based Architecture for Personalized Next Emoji Prediction for Social Media Comments**
D Mistry, JD Plabon, BS Diba, MSH Mukta, MF Mridha

IEEE Access (Q2, Impact Factor: 3.6) – 17 citations

DOI: [10.1109/ACCESS.2024.3448470](https://doi.org/10.1109/ACCESS.2024.3448470) ; Implementation: 

2025 J3 **From centralization to decentralization: Blockchain’s role in transforming social media platforms (Review Article)**

BS Diba, JD Plabon, N Tasnim, M Hossain, D Mistry, S Sarker, MF Mridha, Y Okuyama, J Shin
IEEE Access (Q2, Impact Factor: 3.6) – 10 citations

DOI: [10.1109/ACCESS.2025.3561663](https://doi.org/10.1109/ACCESS.2025.3561663)

J4 **Open problems and challenges in federated learning for IoT: A comprehensive review and strategic guide (Review Article)**

BS Diba, JD Plabon, TJ Mowla, N Nahar, D Mistry, S Sarker, MF Mridha, J Shin
Computers and Electrical Engineering (Q1, Impact Factor: 4.9) – 17 citations

DOI: [10.1016/j.compeleceng.2025.110515](https://doi.org/10.1016/j.compeleceng.2025.110515)

Conference Proceedings

2025 C1 **Personalized Pomodoro Productivity Tracking with Privacy-Preserving Machine Learning in Human-Computer Interaction**

D Mistry, BS Diba, JD Plabon, SU Ahmed, MF Mridha
International Conference on Human-Computer Interaction

DOI: [10.1007/978-3-031-93965-5_29](https://doi.org/10.1007/978-3-031-93965-5_29)

Book Chapters

2025 Ch1 **Privacy-Preserving Kidney Stone Detection from X-Ray Image Using Federated Learning**
JD Plabon, MM Rahaman, MR Tonmoy, MG Kibriya, A Ishraq

Book: Federated Learning in Health Care Technology. **Publisher:** Springer Nature Singapore

DOI: [10.1007/978-981-96-8353-6_4](https://doi.org/10.1007/978-981-96-8353-6_4)

Ch2 **Transforming Healthcare Analytics: Federated Learning for Privacy-Preserving Liver Lesion Detection and Classification**

MS Islam, JD Plabon, N Nahar, N Tasnim, SMRA Masud, J Shin, MR Tonmoy

Book: Federated Learning in Health Care Technology. **Publisher:** Springer Nature Singapore

DOI: [10.1007/978-981-96-8353-6_3](https://doi.org/10.1007/978-981-96-8353-6_3)

Ch3 **Federated Learning Strategies for Confidential Leukemia Detection from Medical Images**
JD Plabon, M Hossain, MA Kabir, D Mistry, MA Hamid

Book: Federated Learning in Health Care Technology. **Publisher:** Springer Nature Singapore

DOI: [10.1007/978-981-96-8353-6_6](https://doi.org/10.1007/978-981-96-8353-6_6)

Interdisciplinary Collaborations

2025 P1 **A Multisectoral Study of Mpox Epidemiology, Resistance Surveillance, and Policy Gaps: Toward a One Health Framework (Review Article)**

D Mistry, M Hossain, BS Diba, JD Plabon, MA Shams, MT Rahman, ASM Bakibillah
Clinical Reviews in Allergy & Immunology (Q1, Impact Factor: 11.3) – 1 citation

DOI: [10.1007/s12016-025-09121-7](https://doi.org/10.1007/s12016-025-09121-7)

Skills

Programming	Python, C++, Java, SQL
Research Areas	Federated Learning, XAI, NLP, LLMs, Deep Learning
Machine Learning	PyTorch, TensorFlow, Hugging Face, Scikit-Learn
Tools	Git, LaTeX, Linux, Docker, VS Code, Jupyter
Web Technologies	Angular, Spring Boot, Django
Languages	Bengali (native), English (professional)

Selected Projects

Gesture Translator

- Developed a real-time ASL-to-English translation system using MediaPipe for hand landmark detection and a custom ML classifier; outputs recognized signs as both text and synthesized audio, enabling accessibility for non-signing users (served as a precursor to the Bangla sign language federated learning research published in J1). 

Presentation Controller

- Built a multimodal hands-free presentation control system combining real-time hand gesture recognition (MediaPipe, OpenCV) with voice command processing (Google Speech API); implements background subtraction and adaptive position prediction for robust performance under varied environments. 

Awards and Achievements

Academic Awards

- **Vice Chancellor's Honor Award** (*X 5: Fall 2023; Spring 2023; Fall 2022; Spring 2022; Spring 2021*)
— An academic award for students who achieved CGPA 3.90+ in a semester.
- **Dean's Honor Award** (*Fall 2021*)
— An academic award for students who achieved CGPA 3.75+ in a semester.

Programming Awards

- **ICPC-Jamilur Reza Choudhury Scholarship Award** (*x3: Fall 2023; Spring 2023; Fall 2022*)
— A prestigious scholarship awarded to top-performing contestants in competitive programming.
- **1st Runner Up**, EEE Tech Fest 2.0 Intra University Programming Contest, 2024
- **2nd Runner Up**, EEE Tech Fest 1.0 Intra University Programming Contest, 2023
- **Champion**, UAP Intra Departmental Hackathon, 2022
- **Champion**, UAP Intra University Programming Contest (Junior Category), 2022

Competitive Programming

- **ICPC Asia Dhaka Regional Contestant**, 2021, 2023, and 2024
- Participated in **10+ National Programming Contests** including NCPC, IUPC (BUET, SUST, IUT, UITS), IUCPC, and more
- Solved **2000+ problems** on online judges including Codeforces, CodeChef, AtCoder, LightOJ, UVA, SPOJ, and CSES
- **Specialist** on Codeforces and **3* Programmer** on CodeChef

Service

Co-Convenor, UAP Programming Contest Club (PCC)

Jul 25 - Present

- Organize and coordinate programming contests, training sessions, workshops and mentor students preparing for ICPC and national-level programming competitions.

President, UAP Programming Contest Club (PCC)

Dec 23 - Jul 24

- Led club operations, organized intra-university programming contests, and mentored members preparing for ICPC and national-level competitions.

President, Robotics Club, CSE Department

Jan 24 - Jul 24

- Directed club activities including robotics workshops, and inter-university competitions.

Volunteer, 45th ICPC World Finals, Dhaka, Bangladesh

Nov 22

- Assisted in operational activities at the most prestigious collegiate programming contest in the world, hosted in Bangladesh for the first time.